

SEMESTER END EXAMINATIONS - MARCH 2024

Program	: B.E. - CSE (AI & ML) / CSE (CY) / AI & ML / AI & DS	Semester	: III
Course Name	: Discrete Mathematical Structures	Max. Marks	: 100
Course Code	: CI35 / CY35 / AI34 / AD34	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Construct the truth tables for the following compound statement CO1 (06)
 $\neg(p \vee (q \wedge r)) \Leftrightarrow (p \vee q) \wedge (p \rightarrow r)$.
 - Prove that the number $\sqrt{2}$ is irrational by method of contradiction. CO1 (07)
 - Using Mathematical induction, prove that for any positive integer n , CO1 (07)
 7 divides $3^{2n+1} + 2^{n+2}$.
- Prove that if $n = ab$ where a, b are positive integers then $a \leq \sqrt{n}$ or CO1 (06)
 $b \leq \sqrt{n}$ by contrapositive method.
 - Without constructing truth tables, prove that CO1 (07)
 $[\neg p \wedge (\neg q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)]$ is logically equivalent to r .
 - Write the converse, inverse and contrapositive for "If a quadrilateral is a CO1 (07)
parallelogram, then its diagonals bisect each other".

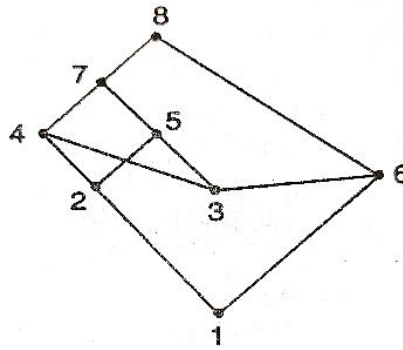
UNIT - II

- Define Boolean Algebra. Verify whether $(S_{42}, /)$ is a Boolean Algebra or CO2 (06)
not?
 - Let $A = \{a, b, c, d\}$ and consider the relation CO2 (07)
 $R = \{(a, a), (a, b), (a, c), (a, d), (b, b), (b, d), (c, c), (c, d), (d, d)\}$. Show that R is a
partial order relation. Hence draw Hasse diagram.
 - Solve the recurrence relation $a_n = a_{n-1} + 2, n \geq 2$ subject to the initial CO2 (07)
condition $a_1 = 3$.
- Let $A = \{1, 2, 3\}$ and R, S be the relations on A whose matrices are as CO2 (06)
given below. Find the relations $\bar{R}, R^{-1}$ and their matrix relations. Also
draw their digraphs.

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad M_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

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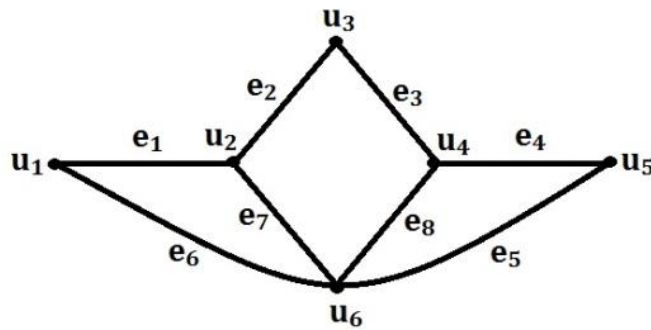
- b) From the following Hasse diagram, find maximal elements, minimal elements, upper bounds, lower bounds, lub and glb of $A = \{2,3\}$ and $B = \{4,6\}$ if exists. CO2 (07)



- c) Solve the recurrence relation $a_n = a_{n-1} + 2a_{n-2}, n \geq 2$ subject to the initial conditions $a_0 = 0, a_1 = 3$. CO2 (07)

UNIT - III

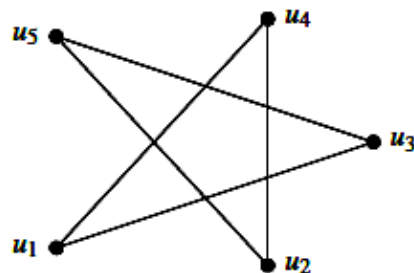
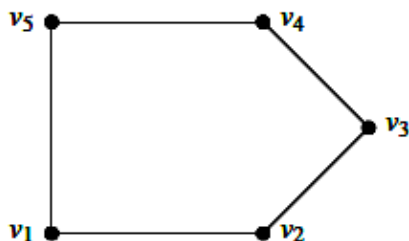
5. a) Define (i) walk (ii) Path (iii) Circuit. CO3 (06)
b) Define fusion of two vertices of a graph. Find (i) Fusion of u_1 and u_2 CO3 (07)
(ii) $G - e_1$ (iii) $G - u_1$ for the graph.



- c) Draw the graph whose incidence matrix is CO3 (07)

$$M = \begin{bmatrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ u & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ v & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ w & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ x & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ y & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

6. a) Define isomorphism of two graphs. Check whether the following graphs are isomorphic or not. CO3 (06)

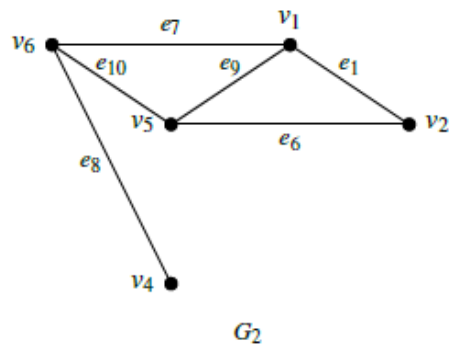
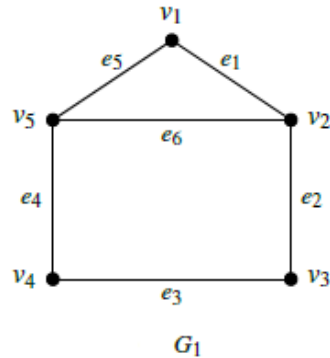


- b) Draw the graph whose adjacency matrix is and hence write its incidence matrix. CO3 (07)

$$X(G) = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

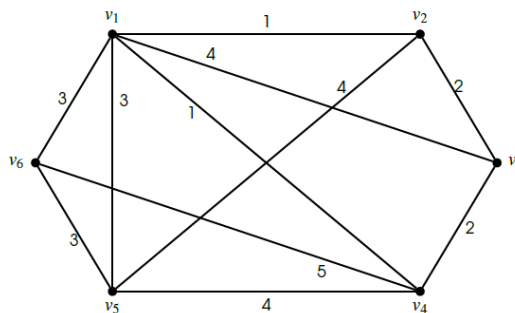
- c) Find $G_1 \cap G_2$, $G_1 \oplus G_2$ of

CO3 (07)

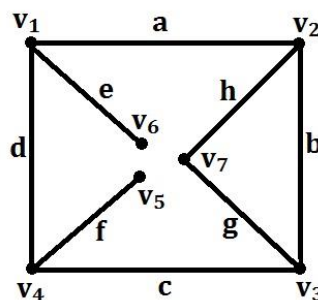


UNIT- IV

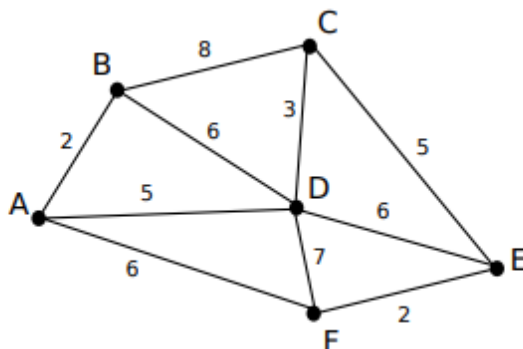
7. a) Define (i) spanning tree (ii) minimally connected graph with an example CO4 (06)
 (iii) cut-sets with an example.
 b) (i) Prove that in a tree, there is one and only path between every pair of CO4 (07)
 vertices.
 (ii) Prove that a tree with n vertices has $n-1$ edges.
 c) Explain Kruskal's algorithm to find the shortest spanning tree and hence CO4 (07)
 find the minimal spanning tree in each of the following graphs.



8. a) Prove that a graph G is a tree if and only if it is minimally connected. CO4 (06)
 b) (i) Define cut-set matrix and weighted graph. CO4 (07)
 (ii) Find cut-sets of the following graph.



- c) Explain Prim's algorithm to find the shortest spanning tree and hence find the minimal spanning tree in each of the following graphs. CO4 (07)

**UNIT - V**

9. a) Define (i) Semi group (ii) Monoid with example for each. CO5 (06)
 b) A non-empty subset H of a group G is a subgroup G if CO5 (07)
 $a, b \in H \Rightarrow ab^{-1} \in H$ where b^{-1} is inverse of b in G .
 c) Let G be a group of positive real numbers under multiplication and G' be CO5 (07)
 a group of all real numbers under addition. Let $f: G \rightarrow G'$ such that
 $f(x) = \log_{10} x$ for $x \in G$. Show that $f: G \rightarrow G'$ is isomorphism
10. a) Define Homomorphism and Isomorphism. CO5 (06)
 If $G = \{1, -1, i, -i\}$ is a group under multiplication then write all subgroups of G .
 b) Let Z^+ be the set of all positive integers and $*$ is an operation defined by CO5 (07)
 $a * b = \text{Max}\{x, y\}$ for $a, b \in Z^+$. Verify $(Z^+, *)$ is Monoid or not.
 c) A non-empty subset H of a group G is a subgroup G if CO5 (07)
 (i) $a \in H, b \in H \Rightarrow ab \in H$
 (ii) $a \in H \Rightarrow a^{-1} \in H$.
